# Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

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## Contents



# 1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise**: how unexpected a contribution is given prior context
- **Transience**: how quickly its influence fades
- **Resonance**: novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,
- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
  - **Human / AI** in Human-AI
  - **Starter-side / Non-starter-side** in same-type conditions

- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

**Primary cross-condition object.** Following the paper's narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as resonance $\sim$ surprise $\times$ slot + (1 | conversation_id) per condition. The cross-condition asymmetry magnitude is

$$A_{\text{influence}}(\text{condition}) \;=\; \left|\beta_{\text{surprise, slot}=1} - \beta_{\text{surprise, slot}=2}\right|,$$

extracted via `emtrends`. Per-story OLS slope differences provide a label-invariant robustness layer. Bootstrap CIs, Cliff's $\delta$ effect sizes, the planned HA vs (HH+AA)/2 contrast, and a length-controlled re-fit are added below. The signed Human/AI decomposition lives in analysis/human-ai/novelty.Rmd and is not duplicated here. Starter-mechanism analysis is deferred.

**Article-use note.** The article-facing result is the magnitude of the novelty-to-resonance slope gap. Do not infer Human-AI specificity from within-condition significance alone; the direct cross-condition question requires comparing the slot-slope gap across conditions and checking the per-story absolute gap as a robustness layer.

# 2 Load Data

Table 1: Loaded novelty data by condition

| condition | n_stories | n_turns |
|---|---|---|
| ai-ai | 80 | 640 |
| human-ai | 96 | 667 |
| human-human | 36 | 288 |

# 3 Preprocessing

```
## Slot-level rows: 3034
```

```
## Metric rows: 9102
```

# 4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: |slope_1 - slope_2| from resonance ~ surprise * slot per condition

| condition | slope_slot1 | slope_slot2 | signed_diff | se | t | p | A_influence | sig |
|---|---|---|---|---|---|---|---|---|
| Human-AI | 0.960 | 0.962 | 0.002 | 0.038 | 0.044 | 0.965 | 0.002 | |
| Human-Human | 1.255 | 0.935 | -0.320 | 0.069 | -4.658 | 0.000 | 0.320 | *** |
| AI-AI | 1.550 | 0.806 | -0.744 | 0.054 | -13.719 | 0.000 | 0.744 | *** |

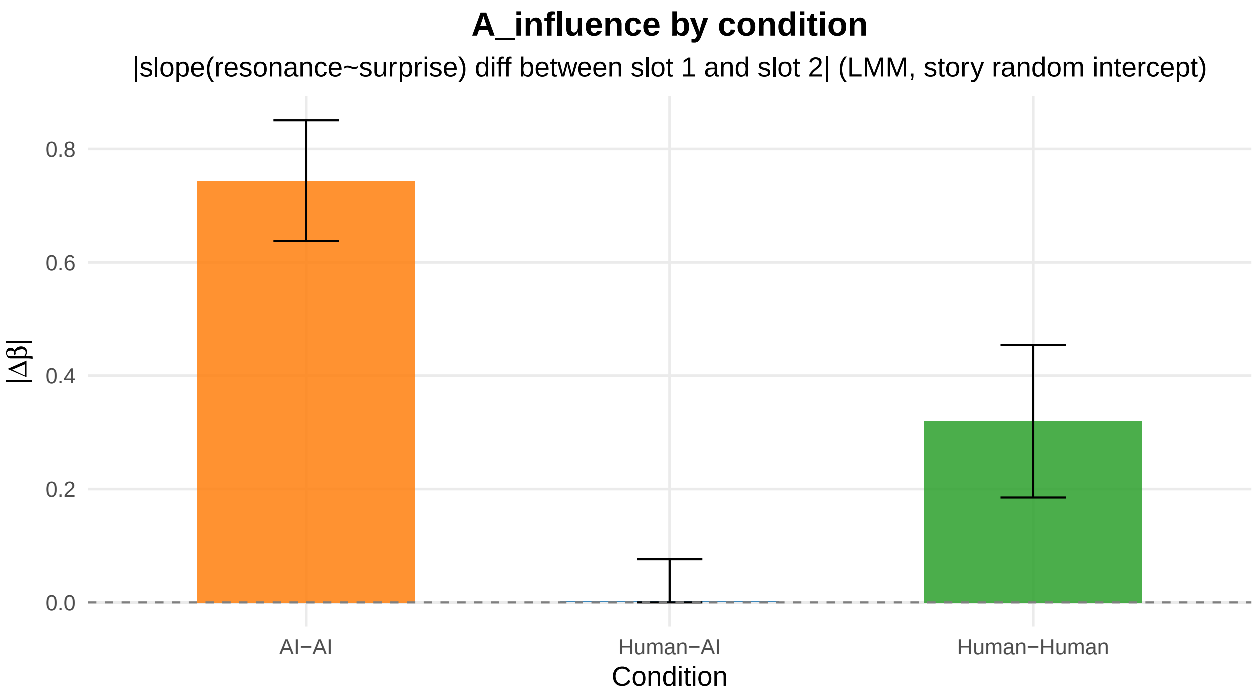



Table 3: Pooled model: |slope_slot1 - slope_slot2| (unsigned) by condition

| condition | A_influence | se |
|---|---|---|
| Human-AI | 0.040 | 0.048 |
| Human-Human | 0.309 | 0.071 |
| AI-AI | 0.753 | 0.046 |

Table 4: Robustness: per-story |slope_1 - slope_2| (OLS per story, 4 points per slot)

| condition | n_stories | median_A | mean_A |
|---|---|---|---|
| Human-AI | 96 | 0.390 | 0.451 |
| Human-Human | 36 | 0.218 | 0.327 |
| AI-AI | 80 | 0.192 | 0.280 |

```
##
## Kruskal-Wallis on per-story A_influence:
##
##  Kruskal-Wallis rank sum test
##
## data:  A_influence_story by condition
## Kruskal-Wallis chi-squared = 14.596, df = 2, p-value = 0.0006769
##
##
## Pairwise Wilcoxon on per-story A_influence (BH-adjusted):
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
```

```
## data:  per_story_slope_diff$A_influence_story and per_story_slope_diff$condition
##
##             Human-AI Human-Human
## Human-Human 0.07968  -
## AI-AI       0.00047  0.40178
##
## P value adjustment method: BH
```

# 5 Bootstrap CIs and Effect Sizes on `A_influence_story`

Table 5: Bootstrap 95% CI on per-condition mean of per-story A_influence (R = 10,000)

| condition | n | mean | ci_lower | ci_upper |
|---|---|---|---|---|
| Human-AI | 96 | 0.451 | 0.386 | 0.522 |
| Human-Human | 36 | 0.327 | 0.242 | 0.421 |
| AI-AI | 80 | 0.280 | 0.226 | 0.338 |

Table 6: Pairwise Cliff's delta on per-story A_influence

| contrast | n_a | n_b | cliff_delta | magnitude |
|---|---|---|---|---|
| Human-AI vs Human-Human | 96 | 36 | 0.219 | small |
| Human-AI vs AI-AI | 96 | 80 | 0.331 | medium |
| Human-Human vs AI-AI | 36 | 80 | 0.098 | negligible |

# 6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: per-story A_influence in HA vs the average of HH and AA

| contrast | estimate | se | t | df | p | ci_lower | ci_upper |
|---|---|---|---|---|---|---|---|
| Human-AI vs (Human-Human + AI-AI)/2 | 0.147 | 0.045 | 3.299 | 157.691 | 0.001 | 0.059 | 0.236 |

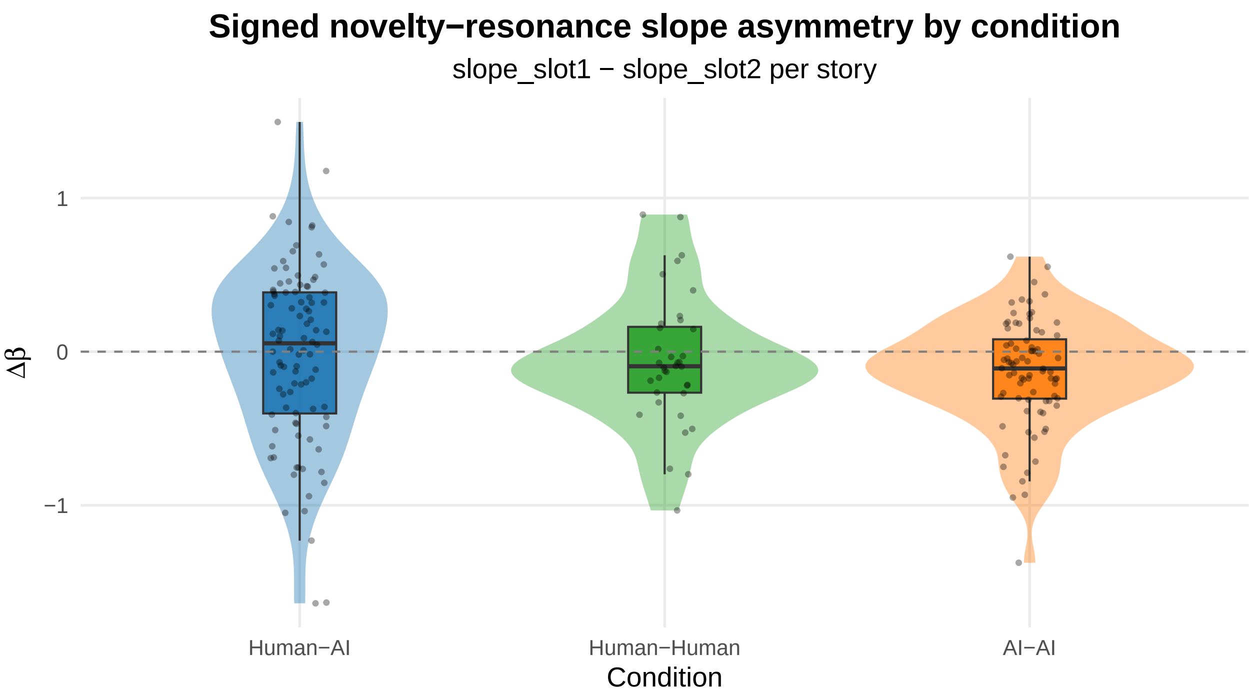
# 7 Directional Stability of Signed Novelty-Resonance Slope Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 novelty-to-resonance slope difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \beta_i^{\text{slot1}} - \beta_i^{\text{slot2}}$. HA-specific identity-tied directionality predicts high stability in HA and low stability in HH/AA, even when unsigned `A_influence` is similar.

Table 8: Per-condition directional stability of signed novelty-resonance asymmetry

| condition | n | mean | sd | stability | sign_consistency | one_sample_t_p |
|---|---|---|---|---|---|---|
| Human-AI | 96 | -0.020 | 0.571 | 0.034 | 0.542 | 0.736 |
| Human-Human | 36 | -0.059 | 0.428 | 0.137 | 0.667 | 0.418 |
| AI-AI | 80 | -0.139 | 0.356 | 0.390 | 0.637 | 0.001 |

```
## Brown-Forsythe variance-equality p = 0.0003
```




# 8 Length-Controlled Variant of `A_influence`

Re-fit the within-condition LMM with turn length as an additive covariate and recompute the unsigned slope-slot magnitude. Tests whether condition-level slot-asymmetry differences survive turn-length adjustment.

Table 9: Length-controlled |slope_slot1 - slope_slot2| (resonance ~ surprise * slot + nwords)

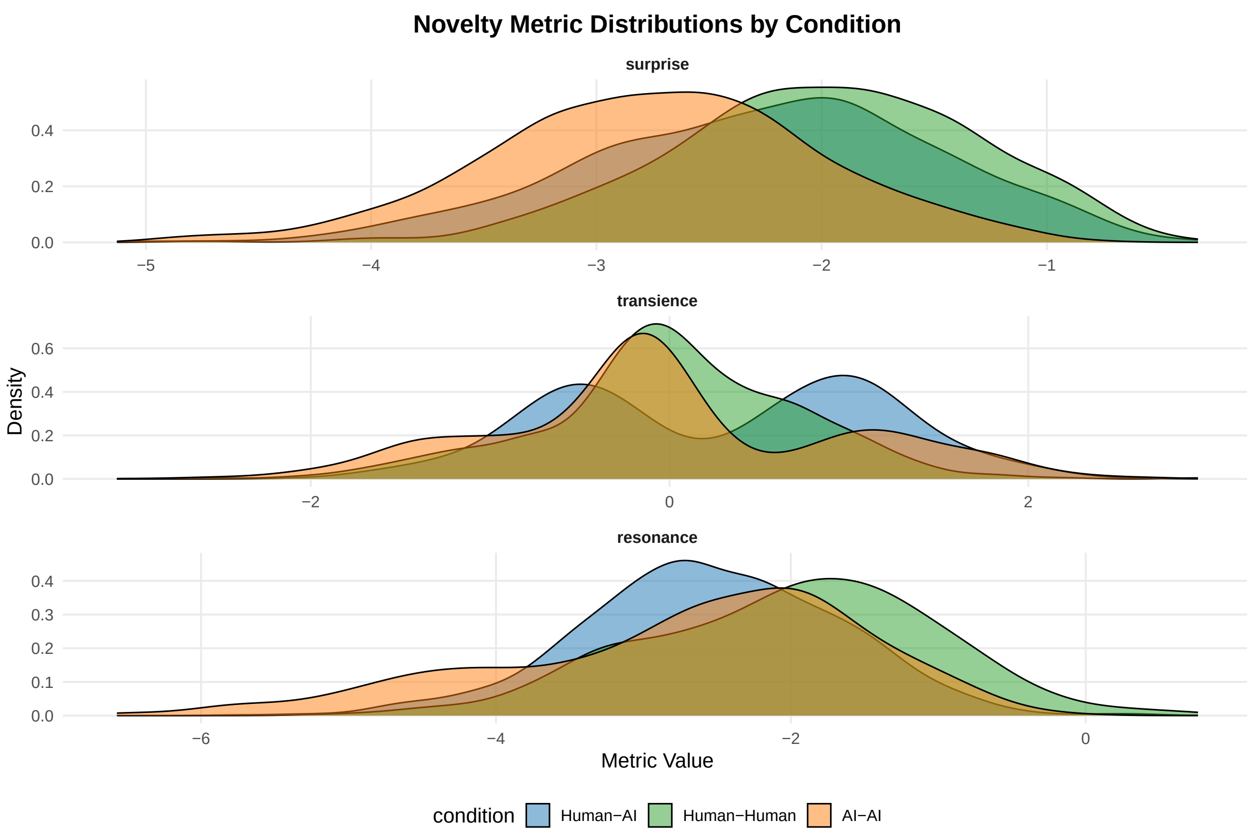
| condition | A_influence_lc | se | p |
|---|---|---|---|
| Human-Human | 0.322 | 0.069 | 0 |

# 9 Supplemental: General Condition-Level Comparison

Table 10: Overall novelty metrics by condition

| condition | metric | n_stories | n_obs | mean_value | sd_value |
|---|---|---|---|---|---|
| Human-AI | surprise | 96 | 1238 | -2.263 | 0.772 |

| condition | metric | n_stories | n_obs | mean_value | sd_value |
|---|---|---|---|---|---|
| Human-AI | transience | 96 | 1238 | 0.283 | 0.901 |
| Human-AI | resonance | 96 | 1238 | -2.546 | 0.869 |
| Human-Human | surprise | 36 | 557 | -1.990 | 0.672 |
| Human-Human | transience | 36 | 557 | 0.016 | 0.718 |
| Human-Human | resonance | 36 | 557 | -2.006 | 0.987 |
| AI-AI | surprise | 80 | 1239 | -2.746 | 0.720 |
| AI-AI | transience | 80 | 1239 | -0.023 | 0.966 |
| AI-AI | resonance | 80 | 1239 | -2.723 | 1.222 |




```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
##    Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 6406.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9667 -0.6209  0.0226  0.6698  2.9961
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.1024   0.3201
```

```
##  Residual                    0.4349   0.6595  
## Number of obs: 3034, groups:  conversation_id, 212
## 
## Fixed effects:
##                       Estimate Std. Error        df t value Pr(>|t|)    
## (Intercept)           -2.26407    0.03767 222.00155 -60.106  < 2e-16 ***
## conditionHuman-Human   0.26796    0.07103 208.87451   3.772 0.000211 ***
## conditionAI-AI        -0.48940    0.05523 212.07380  -8.861 3.25e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.530       
## condtnAI-AI -0.682  0.362
```

```
##  contrast                    estimate     SE  df z.ratio p.value
##  (Human-AI) - (Human-Human)    -0.268 0.0710 Inf  -3.772  0.0005
##  (Human-AI) - (AI-AI)           0.489 0.0552 Inf   8.861  <.0001
##  (Human-Human) - (AI-AI)        0.757 0.0725 Inf  10.444  <.0001
## 
## Degrees-of-freedom method: asymptotic 
## P value adjustment: bonferroni method for 3 tests 
## 
## === transience ===
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
##    Data: filter(df_long, metric == metric_name)
## 
## REML criterion at convergence: 7971.8
## 
## Scaled residuals: 
##     Min      1Q  Median      3Q     Max 
## -3.4041 -0.7094 -0.0559  0.7437  3.2626 
## 
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.0000   0.0000  
##  Residual                    0.8069   0.8983  
## Number of obs: 3034, groups:  conversation_id, 212
## 
## Fixed effects:
##                        Estimate Std. Error         df t value Pr(>|t|)    
## (Intercept)             0.28345    0.02553 3031.00000  11.103  < 2e-16 ***
## conditionHuman-Human   -0.26772    0.04583 3031.00000  -5.842 5.72e-09 ***
## conditionAI-AI         -0.30643    0.03610 3031.00000  -8.489  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.557       
## condtnAI-AI -0.707  0.394
```

```
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
##  contrast                    estimate     SE  df z.ratio p.value
##  (Human-AI) - (Human-Human)   0.2677 0.0458 Inf   5.842  <.0001
##  (Human-AI) - (AI-AI)         0.3064 0.0361 Inf   8.489  <.0001
##  (Human-Human) - (AI-AI)      0.0387 0.0458 Inf   0.845  1.0000
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
##    Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 8749.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.4972 -0.6162  0.0183  0.6623  2.9477
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.1247   0.3531
##  Residual                    0.9706   0.9852
## Number of obs: 3034, groups:  conversation_id, 212
##
## Fixed effects:
##                       Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)           -2.54792    0.04565 229.74015 -55.813  < 2e-16 ***
## conditionHuman-Human   0.53678    0.08539 209.32271   6.286 1.86e-09 ***
## conditionAI-AI        -0.18417    0.06653 214.27113  -2.768  0.00613 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.535
## condtnAI-AI -0.686  0.367
```

```
##  contrast                    estimate     SE  df z.ratio p.value
##  (Human-AI) - (Human-Human)   -0.537 0.0854 Inf  -6.286  <.0001
##  (Human-AI) - (AI-AI)          0.184 0.0665 Inf   2.768  0.0169
##  (Human-Human) - (AI-AI)       0.721 0.0869 Inf   8.297  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
```

# 10 Supplemental: Pooled Novelty-to-Resonance Relationship

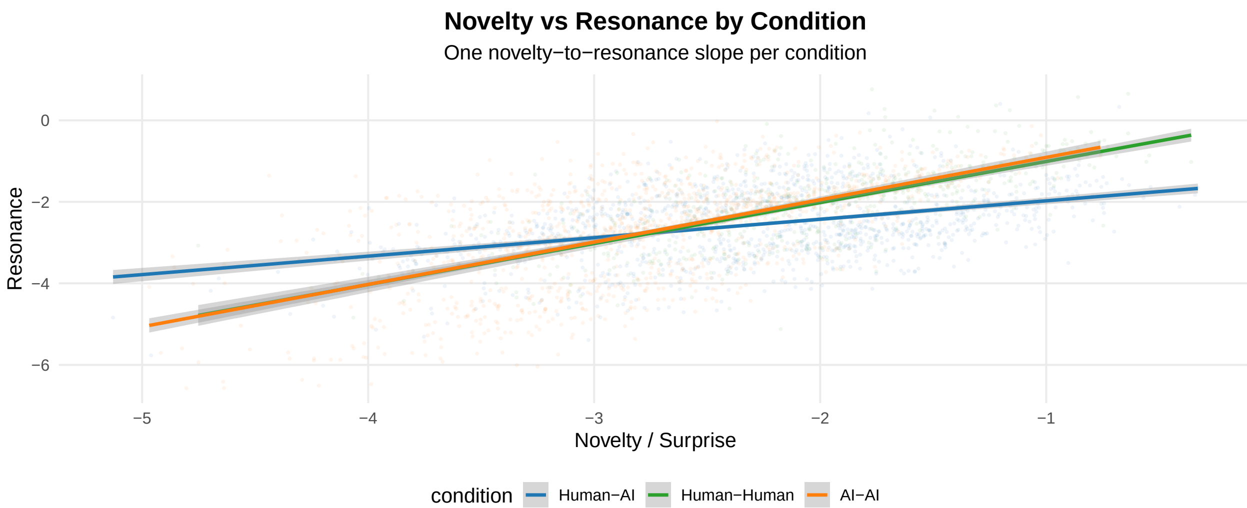
```
## Novelty-to-resonance model:
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

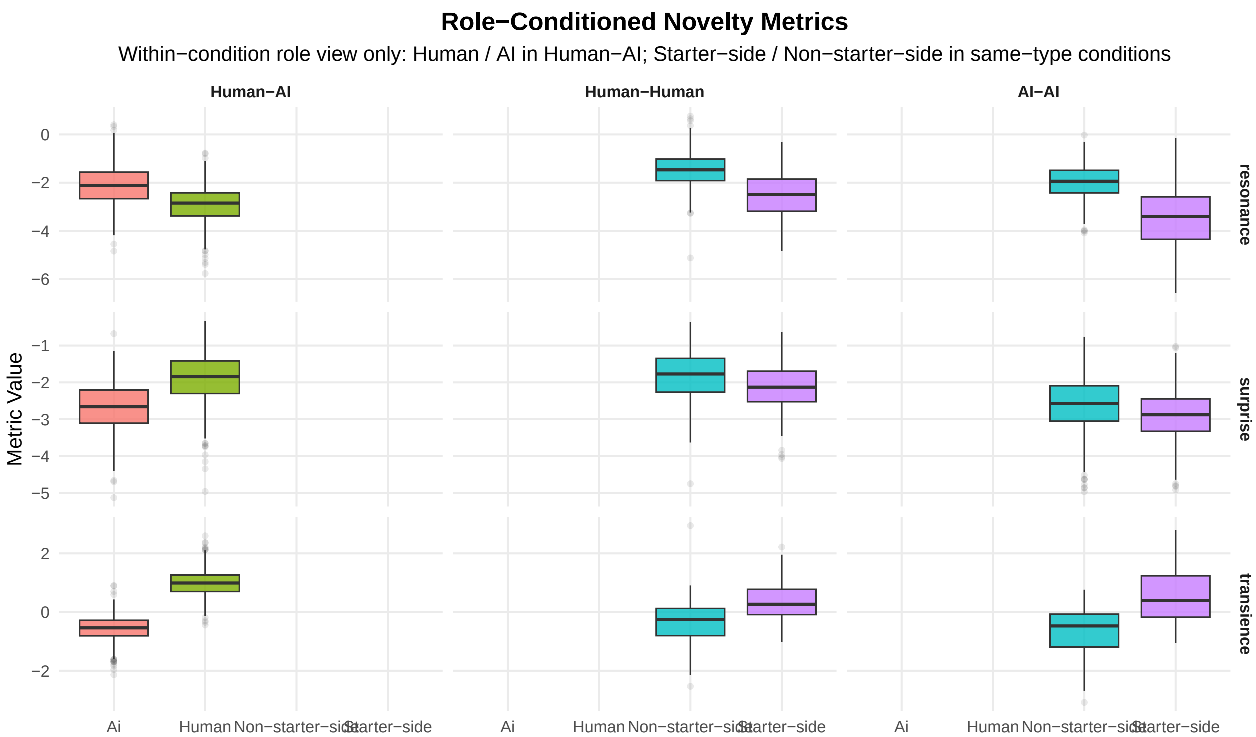
```
## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)
##    Data: df_slots
##
## REML criterion at convergence: 7697.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.4186 -0.6421  0.0480  0.6002  3.6450
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.0000   0.0000
##  Residual                    0.7343   0.8569
## Number of obs: 3034, groups:  conversation_id, 212
##
## Fixed effects:
##                               Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                   -1.52224    0.07543 3028.00000  -20.18   <2e-16
## surprise                       0.45251    0.03155 3028.00000   14.34   <2e-16
## conditionHuman-Human           1.51953    0.13635 3028.00000   11.14   <2e-16
## conditionAI-AI                 1.65192    0.12208 3028.00000   13.53   <2e-16
## surprise:conditionHuman-Human  0.55403    0.06260 3028.00000    8.85   <2e-16
## surprise:conditionAI-AI        0.58634    0.04624 3028.00000   12.68   <2e-16
##
## (Intercept)                   ***
## surprise                      ***
## conditionHuman-Human          ***
## conditionAI-AI                ***
## surprise:conditionHuman-Human ***
## surprise:conditionAI-AI       ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise     0.946
## cndtnHmn-Hm -0.553 -0.524
## condtnAI-AI -0.618 -0.585  0.342
## srprs:cnH-H -0.477 -0.504  0.946  0.295
## srprs:AI-AI -0.646 -0.682  0.357  0.955  0.344
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
##
## Condition-specific novelty-to-resonance slopes:
```

```
##  condition   surprise.trend     SE  df asymp.LCL asymp.UCL
##  Human-AI             0.453 0.0315 Inf     0.391     0.514
##  Human-Human          1.007 0.0541 Inf     0.901     1.113
##  AI-AI                1.039 0.0338 Inf     0.973     1.105
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```




## 11 Supplemental: Role-Level View




## 12 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|$ from `resonance ~ surprise * slot + (1|conversation_id)`. The pooled LMM contributes only the unsigned magnitude column. Per-story OLS slope differences provide the label-invariant robustness test.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of per-story `A_influence`
- pairwise Cliff's $\delta$ effect sizes
- planned single-df contrast HA vs (HH+AA)/2

- length-controlled re-fit (`+ nwords`) of the unsigned magnitude

Supplemental views retained:

- condition-level marginal distributions and pooled `condition` LMMs
- pooled novelty-to-resonance model with `surprise * condition` interaction
- within-condition role box-plots (descriptive only; HA Human/AI inferential view lives in analysis/human-ai/novelty.Rmd)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

*Analysis completed: 2026-05-10 20:39:36.278611*